STAT 5703 **Name & UNI: Solution**

Spring 2026

Midterm

Instructions:

1. There are 5 questions on this test.
2. Show all work.
3. Approved: One-page one-sided formula sheet, any calculator (without WiFi).
4. Tables: Attached.
5. If α is not provided, then the hypothesis test must be done at the 5% level.
6. Do not stay too long at a part where you experience a lot of difficulties.

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1. True or False: (1p each)

   a) If for two sequences of random variables we have that $\boldsymbol{X_n} \xrightarrow{d} \boldsymbol{X}$ and $\boldsymbol{Y_n} \xrightarrow{d} \boldsymbol{Y}$, where X and Y are random variables, then $\boldsymbol{X_n Y_n} \xrightarrow{d} \boldsymbol{XY}$. F

   b) All MLE estimators are guaranteed to be unbiased. F

   c) In cross-validation we use the training data set to build many candidate models and select the model that gives the lowest prediction error when applied to the test data set. T

   d) The Neyman-Pearson approach is used when testing a simple vs. simple hypothesis. T

   e) An estimator is consistent if its variance converges to 0 as the sample size increases. F

   f) The power of a test, for any value of θ, is equal to one minus the probability of Type I error. F

   g) The power curve is always an increasing function of the parameter. F

   h) Suppose we obtain one bootstrap sample from the following data:

   1, 3, 7, 5, 5, 9, 2, 8, 8, 2

   Then the probability that at least one 8 appears in the bootstrap sample is approx. 0.89 T

   i) Controlling for a collider (a variable caused by both the treatment and outcome) can create a spurious association between them. T

   j) Regression step functions are continuous at the knots. F

   k) *Bonus* question:

Suppose that the vector $\boldsymbol{X} \sim N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu} = \binom{1}{0}, \boldsymbol{\Sigma} = \begin{pmatrix} 1 & -0.2 \\ -0.2 & 1 \end{pmatrix}$.

Then there is 95% probability that $\boldsymbol{X}$ is within an ellipse that can be expressed as $(X_1 - 1)^2 - 0.4(X_1 - 1)X_2 + X_2^2 \leq 5.991$ F

2. In this problem we want to understand how price affects the demand for cheese. In the data we used each row correspond to a week of sales for a particular Kroger store. There were 706 stores sampled in 11 cities. For each store three variables were recorded: `city` represents the city name of where the store is located, while the `price` and `vol` represent price (in $ per lb) and sales volume (in number of packages of cheese sold). We fit two linear regression models to predict the volume by the other variables. The resulting `R` code is presented below:

```
> m1 = lm(vol ~ price, data = kroger)
> summary(m1)
lm(formula = vol ~ price, data = kroger)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   9363.2      708.2  13.221  < 2e-16 ***
price        -1738.4      265.9  -6.537  1.2e-10 ***
---

Residual standard error: 2415 on 704 degrees of freedom
Multiple R-squared:  0.05723,                Adjusted R-squared:  0.05589
F-statistic: 42.74 on 1 and 704 DF,  p-value: 1.202e-10

> m2 = lm(vol ~ price + city, data=kroger)
> summary(m2)
lm(formula = vol ~ price + city, data = kroger)

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)       14950.9      494.2  30.250  < 2e-16 ***
price             -3491.4      164.7 -21.200  < 2e-16 ***
cityBirmingham    -4707.1      239.2 -19.676  < 2e-16 ***
cityCincinnati      825.3      234.4   3.521 0.000458 ***
cityColumbus      -1224.3      236.9  -5.168 3.10e-07 ***
cityDallas         -297.0      234.9  -1.264 0.206490
cityDetroit        -430.2      227.9  -1.887 0.059518 .
cityHouston        4074.5      236.0  17.267  < 2e-16 ***
cityIndianapolis  -1725.9      230.7  -7.480 2.26e-13 ***
cityLouisville    -2581.3      237.3 -10.876  < 2e-16 ***
cityNashville     -2650.3      238.6 -11.107  < 2e-16 ***
cityRoanoke       -1142.2      233.5  -4.892 1.24e-06 ***
---
Residual standard error: 1292 on 694 degrees of freedom
Multiple R-squared:  0.7337,                  Adjusted R-squared:  0.7295
F-statistic: 173.9 on 11 and 694 DF,  p-value: < 2.2e-16
```

a) (4 p) Looking at models 1 and 2, choose *all* correct statements:

i. Volume drops by 1738.4 packages on average for every $1 increase in price.
ii. Volume drops by 3491.4 packages on average for every $1 increase in price.
iii. Volume drops by 3491.4 packages on average for every $1 increase in price, in a fixed city.
iv. The total effect of price is not a significant predictor at $\alpha = 0.05$ level.

b) (2 p) Give the *most likely* explanation of the discrepancies of the price effect between models 1 and 2.

i. There is a programming error.
ii. The city is a much better predictor of volume which resulted in much lower overall residual standard error allowing the price coefficient to became larger.
iii. Both models overfit the data.
iv. There is a fork or pipe confounding effect between price, city and volume.

c) (4 p) Choose all correct statements about model 2:

i. The average volume of packages sold is 14950.9.
ii. This is a model that fits parallel lines.
iii. This is known as a model with interaction effects.
iv. This is a model that fits non-parallel lines.

d) (4 p) Using model 2, write down the fitted equation for Houston:

Volume for Houston = 14950.9 + 4074.5 - 3491.4*Price = 19025.4 - 3491.4*Price

e) (4 p) Output for the 5 largest hat values and 5 largest absolute studentized residuals is provided:

```
> sort(hatdiag, dec = T)[1:5]
       243        463        339         20        130
0.03925730 0.03737099 0.03219400 0.02907258 0.02802711

> sort(abs(rstudent(m2)), dec = T)[1:5]
      42      310      361      368       75
5.466336 4.589955 4.577160 4.367294 4.054870
```

Choose *all* correct statements:

i. There are more than two outliers.
ii. There are exactly two leverage points.
iii. There are exactly two influential points.
iv. There are exactly two outliers.

f) (2 p) Which plot is the best if you wanted to check the linearity assumption of model 1:

i. Histogram of the residuals.
ii. Residuals vs fitted values.
iii. Q-Q plot of the residuals.
iv. Boxplot of the residuals

3. Let $X_1, \dots, X_n$ be a random sample from the Geometric distribution with pmf

$$p(x;\theta) = \theta(1-\theta)^{x-1}, x = 1,2,\dots,$$

where $\theta \in (0,1)$ is an unknown parameter.

a) (7 p) Find the method of moments estimator of θ.

Note: No need to derive the mean of the Geometric, as it should be on their formula sheet.

$$E(X) = \frac{1}{\theta}$$

$$\frac{1}{\theta} = \bar{X} \Rightarrow \hat{\theta}_{MM} = \frac{1}{\bar{X}}$$

b) (8 p) Find the MLE estimator of θ.

$$L(\theta) = \prod_{i=1}^{n} [\theta(1-\theta)^{x_i-1}] = \theta^n (1-\theta)^{\sum_{i=1}^n x_i - n}$$

$$\Rightarrow l(\theta) = n\log\theta + \left(\sum_{i=1}^{n} x_i - n\right)\log(1-\theta)$$

$$\Rightarrow s(\theta) = \frac{n}{\theta} - \frac{\sum_{i=1}^n x_i - n}{1-\theta}$$

$$\Rightarrow s(\theta) = 0 \iff \frac{n}{\theta} - \frac{\sum_{i=1}^n x_i - n}{1-\theta} = 0$$

$$\iff n(1-\theta) = \left(\sum_{i=1}^{n} x_i - n\right)\theta$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{\bar{X}}$$

2p for *L*, 2p for *l*, 2p for *s* and 2p for solving *s* = 0

c) (8 p) Derive the Fisher information $I_n(\theta)$ and state the asymptotic distribution of the MLE.

$$l''(\theta) = \frac{\partial}{\partial\theta}\left(\frac{n}{\theta} - \frac{\sum_{i=1}^n X_i - n}{1-\theta}\right) = -\frac{n}{\theta^2} - \frac{\sum_{i=1}^n X_i - n}{(1-\theta)^2}$$

$$\Rightarrow I_n(\theta) = E\left[\frac{n}{\theta^2} - \frac{\sum_{i=1}^n X_i - n}{(1-\theta)^2}\right] = \frac{n}{\theta^2} + \frac{E(\sum_{i=1}^n X_i) - n}{(1-\theta)^2} = \frac{n}{\theta^2} + \frac{\frac{n}{\theta} - n}{(1-\theta)^2}$$

$$= \frac{n}{\theta}\left(\frac{1}{\theta} + \frac{1}{1-\theta}\right) = \frac{n}{\theta^2(1-\theta)}$$

$$\Rightarrow \frac{\frac{1}{\bar{X}} - \theta}{\sqrt{\frac{\theta^2(1-\theta)}{n}}} \xrightarrow{d} N(0,1)$$

Alternatively, $\sqrt{n}\left(\frac{1}{\bar{X}} - \theta\right) \xrightarrow{d} N(0, \sqrt{\theta^2(1-\theta)})$ or $\frac{\sqrt{n}}{\bar{X}} \xrightarrow{d} N(\theta, \sqrt{\theta^2(1-\theta)})$

There shouldn't be any *n* in the normal limit!

Deduct -2 points for answers with *n* in the limit.

4. Let $X_1, \dots, X_n$ be random sample from Bernoulli(p) with pmf $f(x) = p^x(1-p)^{1-x}$.

a. (8 p) Derive the form of the Neyman-Pearson rejection region for the most powerful test of

$$H_0: p = \frac{3}{5} \; vs. \; H_a: p = \frac{2}{5}$$

$$L(p) = \prod_{i=1}^{n} [p^{x_i}(1-p)^{1-x_i}] = p^{\sum_{i=1}^{n} x_i}(1-p)^{n-\sum_{i=1}^{n} x_i}$$

$$RR = \frac{L(p_a)}{L(p_0)} \geq k \Leftrightarrow \frac{\left(\frac{2}{5}\right)^{\sum_{i=1}^{n} x_i}\left(\frac{3}{5}\right)^{n-\sum_{i=1}^{n} x_i}}{\left(\frac{3}{5}\right)^{\sum_{i=1}^{n} x_i}\left(\frac{2}{5}\right)^{n-\sum_{i=1}^{n} x_i}} \geq k$$

$$\Leftrightarrow \left(\frac{2}{3}\right)^{\sum_{i=1}^{n} x_i}\left(\frac{3}{2}\right)^{n-\sum_{i=1}^{n} x_i} \geq k_1$$

$$\Leftrightarrow \left(\frac{2}{3}\right)^{2\sum_{i=1}^{n} x_i - n} \geq k_1$$

$$\Leftrightarrow \sum_{i=1}^{n} x_i \leq c$$

b. (7 p) Regardless of your answer to part a), suppose n = 8 and we want to use the rejection region $RR = \{\sum_{i=1}^{n} X_i \leq 1\}$. Calculate the Type I and Type II errors.

Let $X = \sum_{i=1}^{n} X_i \Rightarrow X \sim Binom(8, p)$

$$\Rightarrow \alpha = P\left(X \leq 1 | p = \frac{3}{5}\right) = \binom{8}{0}\left(\frac{3}{5}\right)^0\left(\frac{2}{5}\right)^8 + \binom{8}{1}\left(\frac{3}{5}\right)^1\left(\frac{2}{5}\right)^7 = 0.0085$$

$$\beta\left(\frac{2}{5}\right) = 1 - P\left(X \leq 1 | p = \frac{2}{5}\right)$$

$$= 1 - \binom{8}{0}\left(\frac{2}{5}\right)^0\left(\frac{3}{5}\right)^8 - \binom{8}{1}\left(\frac{2}{5}\right)^1\left(\frac{3}{5}\right)^7 = 0.89$$

c. (8 p) Use α = 0.05 and the *asymptotic* rejection rule to perform the likelihood ratio test of

$$H_0: p = 0.6 \; vs. \; H_a: p \neq 0.6$$

Use without proof that $\hat{p}_{MLE} = \frac{\sum_{i=1}^{n} X_i}{n}$ and perform the test on data with $\sum_{i=1}^{n} X_i = 16, n = 30$

$$L(p) = p^{\sum_{i=1}^{n} x_i}(1-p)^{n-\sum_{i=1}^{n} x_i}$$

$$\Rightarrow l(p) = \sum_{i=1}^{n} x_i \log p + \left(n - \sum_{i=1}^{n} x_i\right)\log(1-p) = 16\log(p) + 14\log(1-p)$$

$$\Rightarrow -2\log\Lambda(X) = -2[l(p_0) - l(\hat{p}_{MLE})]$$

$$= -2\left[16\log(0.6) + 14\log(0.4) - 16\log\left(\frac{16}{30}\right) - 14\log\left(\frac{14}{30}\right)\right] = 0.547162$$

Df = 1, $\chi^2_{crit} = 3.841$

Since 0.547 < 3.841 we fail to reject H_0

5. Let $X_1, X_2, \ldots, X_n \sim Exp(\theta)$ with pdf $f(x) = \theta e^{-\theta x}, x > 0$, where $\theta > 0$ is an unknown parameter. Use without proving it the fact that for $Exp(\theta)$ the MLE is $\hat{\theta}_{MLE} = \frac{1}{\bar{X}}$

a) (8 p) Construct a 95% *asymptotic* CI for the parameter θ based on the MLE and a sample of 30 observations which resulted in $\hat{\theta}_{MLE} = \frac{1}{\bar{X}} = 0.5671$

By CLT we have

$$\frac{\bar{X} - \frac{1}{\theta}}{\sqrt{\frac{1}{n\theta^2}}} \xrightarrow{d} N(0,1) \Leftrightarrow \frac{\sqrt{n}\left(\bar{X} - \frac{1}{\theta}\right)}{\frac{1}{\theta}} \xrightarrow{d} N(0,1)$$

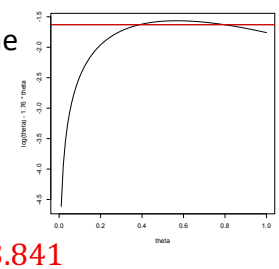
By the delta method, with $g(x) = \frac{1}{x}, g'(x) = -\frac{1}{x^2}$ we have that

$$\frac{\sqrt{n}\left(g(\bar{X}) - g\left(\frac{1}{\theta}\right)\right)}{g'\left(\frac{1}{\theta}\right)\frac{1}{\theta}} \xrightarrow{d} N(0,1) \Leftrightarrow \frac{\sqrt{n}\left(\frac{1}{\bar{X}} - \theta\right)}{\frac{\theta^2}{\theta}} \xrightarrow{d} N(0,1) \Leftrightarrow \frac{\sqrt{n}\left(\frac{1}{\bar{X}} - \theta\right)}{\theta} \xrightarrow{d} N(0,1)$$

Therefore, a 95% *asymptotic* CI for the parameter θ is

$$\frac{1}{\bar{X}} \pm 1.96\frac{\hat{\theta}}{\sqrt{n}} = 0.5671 \pm 1.96\left(\frac{0.5671}{\sqrt{30}}\right) = (0.36, 0.77)$$

b) (8 p) Derive the defining inequality of the profile CI for θ. (Do not try to solve the inequality).



$$L(\theta) = \prod_{i=1}^{n}\left(\theta e^{-\theta x_i}\right) = \theta^n e^{-\theta \sum_{i=1}^{n} x_i} \Rightarrow l(\theta) = n\log\theta - \theta\sum_{i=1}^{n} x_i$$

$$\Rightarrow 2l_n(\hat{\theta}) - 2l_n(\theta) < \chi^2_\alpha \Leftrightarrow 2\left(n\log\hat{\theta} - \hat{\theta}\sum_{i=1}^{n} x_i - n\log\theta + \theta\sum_{i=1}^{n} x_i\right) < 3.841$$

That is, 95% profile CI consists of

$$\{\theta: 30 * \log 0.5671 - 0.5671 * 52.9 - 30\log\theta + 52.9\theta < 1.92\}$$
$$\Leftrightarrow \{\theta: -30\log\theta + 52.9\theta < 48.936\} \Leftrightarrow \{\theta: \log\theta - 1.76\theta > -1.63\}$$

c) (8 p) Define an estimator $\hat{\theta} = \min\{X_1, \ldots, X_n\}$. You may use without proof the fact that the distribution of $\hat{\theta}$ is $\hat{\theta} \sim Exp(n\theta)$, which means $\theta\hat{\theta} \sim Exp(n)$ is a pivot. Construct a 95% CI for the parameter θ using the above pivot and a sample with n = 30 which resulted in $\hat{\theta} = 0.2$.

Hint: You may need here the inverse cdf of $Exp(n)$ which is $F^{-1}(p) = -\frac{\ln(1-p)}{n}, p \in (0,1)$.

First find the quantiles: $\exp_{0.025} = F^{-1}(0.025) = -\frac{\ln(0.975)}{30} = 0.000843927$ and $\exp_{0.975} = F^{-1}(0.975) = -\frac{\ln(0.025)}{30} = 0.1229626$

Then we have that

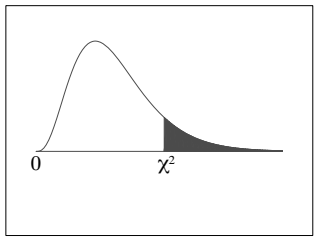
$$P(0.00084393 < \theta\hat{\theta} < 0.1229626) = 0.95$$
$$\Rightarrow P\left(\frac{0.00084393}{\hat{\theta}} < \theta < \frac{0.1229626}{\hat{\theta}}\right) = 0.95$$

A 95% CI for parameter θ is $\left(\frac{0.00084393}{0.2} < \theta < \frac{0.1229626}{0.2}\right) = (0.00421965, 0.614813)$

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	**0.50**	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.01**	**0.005**	**0.001**	**0.0005**
two-tails	**1.00**	**0.50**	**0.40**	**0.30**	**0.20**	**0.10**	**0.05**	**0.02**	**0.01**	**0.002**	**0.001**
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_\alpha$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169